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# BATCH

In the book Lean Thinking, James Womack and Daniel Jones recount a story of stuffing newsletters into envelopes with the assistance of one of the author's two young children. Every envelope had to be addressed, stamped, filled with a letter, and sealed. The daughters, age six and nine, knew how they should go about completing the project: "Daddy, first you should fold all of the newsletters. Then you should attach the seal. Then you should put on the stamps." Their father wanted to do it the counterintuitive way: complete each envelope one at a time. They —like most of us—thought that was backward, explaining to him "that wouldn't be efficient!" He and his daughters each took half the envelopes and competed to see who would finish first.

The father won the race, and not just because he is an adult. It happened because the one envelope at a time approach is a faster way of getting the job done even though it seems inefficient. This has been confirmed in many studies, including one that was recorded on video.[1]

The one envelope at a time approach is called "single-piece flow" in lean manufacturing. It works because of the surprising power of small batches. When we do work that proceeds in stages, the "batch size" refers to how much work moves from one stage to the next at a time. For example, if we were stuffing one hundred envelopes, the intuitive way to do it—folding one hundred letters at a time—would have a batch size of one hundred. Single-piece flow is so named because it has a batch size of one.